

RAMBO: RL-Augmented Model-Based Whole-Body Control for Loco-Manipulation

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website



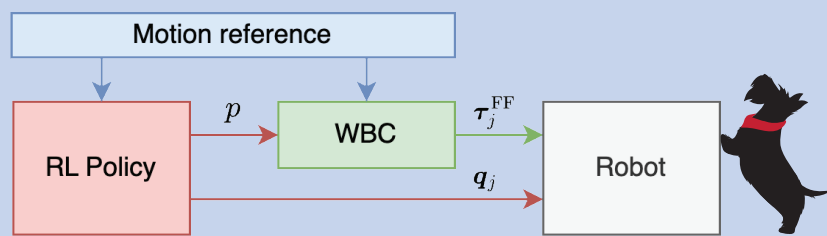
video



code

1 Introduction

Loco-manipulation, physical interaction of objects that is concurrently coordinated with locomotion, remains a major challenge for legged robots due to the need for both precise end-effector control and robustness to unmodeled dynamics. We introduce RAMBO, a hybrid framework that integrates model-based whole-body control (WBC) within a feedback policy trained with reinforcement learning (RL).



2 Method Overview

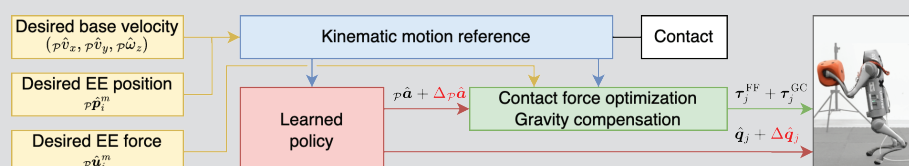
- Motion reference is generated from target base velocity, gait pattern, target end-effector position.
- The target base acceleration is calculated from a PD controller. Together with target end-effector force, they are sent to the whole-body controller using Single Rigid Body (SRB) model to optimize reaction forces via a quadratic program (QP).

$$\min_{\mathcal{B}\mathbf{u}_i, i \in \mathcal{N}} \|\Delta \mathcal{B}\mathbf{a}_U\|^2 + \sum_{i \in \mathcal{M}} \|\Delta \mathcal{B}\mathbf{u}_{iV}\|^2 + \sum_{i \in \mathcal{L}} \|\mathcal{B}\mathbf{u}_{iW}\|^2$$

$$\text{subject to : } \mathcal{B}\mathbf{a} = \sum_{i=1}^N \mathbf{A}_i \mathcal{B}\mathbf{u}_i + \hat{\mathbf{g}}$$

$$\begin{aligned} (\mathcal{B}\mathbf{u}_i)_z &= 0, & i &\in \mathcal{L}_{\text{swing}} \\ \|(\mathcal{B}\mathbf{u}_i)_x\| &\leq \mu(\mathcal{B}\mathbf{u}_i)_z, & i &\in \mathcal{L}_{\text{stance}} \\ \|(\mathcal{B}\mathbf{u}_i)_y\| &\leq \mu(\mathcal{B}\mathbf{u}_i)_z, & i &\in \mathcal{L}_{\text{stance}} \\ u_{z,\min} &\leq (\mathcal{B}\mathbf{u}_i)_z \leq u_{z,\max} & i &\in \mathcal{L}_{\text{stance}} \end{aligned}$$

- An RL policy is trained to provide corrective feedback to base acceleration and joint position targets.



- The total joint torque combines a feedforward term from WBC, a gravity compensation term, and a PD term from target joint positions.

$$\tau_j = \tau_j^{\text{FF}} + \tau_j^{\text{GC}} + k_p(\tilde{q}_j - q_j) + k_d(\dot{\tilde{q}}_j - \dot{q}_j)$$

Some other details

Simulation: Isaac Lab [1]
QP solver: qpnh [2] (train), OSQP [3] (deploy)
Control frequency: 100 Hz
Action scale: 5.0 (base head), 0.15 (joint head)
Joint stiffness: 40
Joint damping: 1.0

Max velocity: ± 0.5 m/s (forward),
 $\pm 0.5/0$ m/s (sideway), ± 0.5 rad/s (turn)
End-effector position: $[0.19, 0.5]/[0.15, 0.30]$ m (x),
 $\pm[0, 0.2]$ m (y), $[0.0, 0.4]/[0.3, 0.9]$ m (z)
Max end-effector force: $\pm 30/20$ N
(quadruped/biped)

3 Results and Discussion

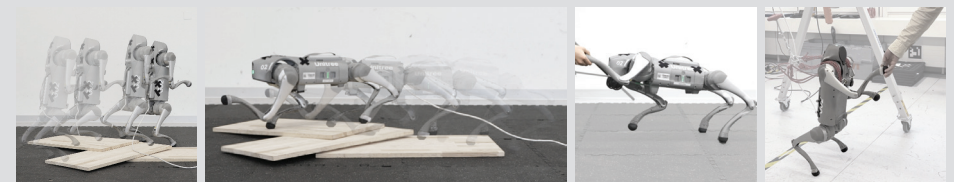
We demonstrate RAMBO on a range of loco-manipulation tasks in both bipedal and quadrupedal scenarios.



Precision

	Error in tracking from quadruped task			Error in tracking from biped task		
	lin vel (m/s) ↓	ang vel (rad/s) ↓	EE pos (m) ↓	lin vel (m/s) ↓	ang vel (rad/s) ↓	EE pos (m) ↓
Vanilla	0.387 ± 0.022	0.257 ± 0.026	0.254 ± 0.009	0.313 ± 0.026	0.353 ± 0.066	0.431 ± 0.024
Imitation	0.383 ± 0.022	0.253 ± 0.025	0.257 ± 0.010	0.310 ± 0.027	0.334 ± 0.060	0.448 ± 0.018
Residual	0.153 ± 0.027	0.153 ± 0.067	0.077 ± 0.013	0.383 ± 0.054	0.508 ± 0.271	0.347 ± 0.030
RAMBO-base	0.321 ± 0.041	0.293 ± 0.126	0.168 ± 0.036	0.305 ± 0.060	0.389 ± 0.171	0.453 ± 0.025
RAMBO-joint	0.108 ± 0.014	0.101 ± 0.032	0.046 ± 0.007	0.306 ± 0.046	0.383 ± 0.109	0.134 ± 0.044
RAMBO-ff	0.374 ± 0.036	0.404 ± 0.154	0.286 ± 0.048	0.320 ± 0.061	0.389 ± 0.187	0.447 ± 0.026
RAMBO	0.087 ± 0.009	0.085 ± 0.022	0.039 ± 0.003	0.286 ± 0.039	0.352 ± 0.075	0.036 ± 0.002

Robustness and compliance



4 Conclusion

- We present RAMBO, a hybrid control framework that combines model-based whole-body control with an RL policy to enable robust and precise loco-manipulation.
- RAMBO provides flexible trade-off tracking performance for compliance, providing safe and adaptive interactions with the environment.

References

- [1] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar et al., "Orbit: A unified simulation framework for interactive robot learning environments," IEEE Robotics and Automation Letters, vol. 8, no. 6, pp. 3740–3747, 2023.
- [2] B. Amos and J. Z. Kolter, "Optnet: Differentiable optimization as a layer in neural networks," in International conference on machine learning. PMLR, 2017, pp. 136–145.
- [3] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "Osqp: An operator splitting solver for quadratic programs," Mathematical Programming Computation, vol. 12, no. 4, pp. 637–672, 2020.